

A Countable Coprime Renewal Determinant: Trace Class, Exact Cycle Ledgers, and the Operator Boundary

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Abstract

We freeze a target-free countable renewal suspension whose symbols are integers $n \geq 2$ and whose adjacent symbols are required to be coprime. The roof is $\log n$, and a symmetric half-roof convention gives the transfer kernel

$$(L_s)_{mn} = \mathbf{1}_{\gcd(m,n)=1} (mn)^{-s/2}.$$

For $\Re s > 1$, a Möbius rank-one expansion proves local uniform trace-norm convergence, holomorphicity of the family, and existence of the same-object Fredholm determinant $D_{\text{cop}}(s) = \det_F(I - L_s)$. Finite projections then give an exact cyclic trace-power formula and the primitive-repetition ledger. We certify the absence of period-one cycles, the orientation factors for periods two and three, and rational finite cutoffs through repetition power six. The boundary is sharp for the frozen ℓ^2 realization: the e_2 column is not square summable when $\Re s \leq 1$. These results establish an analytic Route-A theorem edge, not a prime-orbit law or a completed- ξ determinant. We state the remaining continuation question without evaluating determinant roots.

1 Introduction

The finite-state symbolic models examined in the preceding search were useful controls, but their finite exponential structure forces an $O(T)$ divisor count. A different object must therefore be infinite-dimensional before a global determinant comparison is even meaningful. This paper records the first audit of one such object: a countable shift with a local coprimality constraint and an intrinsic logarithmic roof.

The construction is deliberately modest. It asks whether the transfer operator is mathematically defined, whether its traces have a reproducible primitive-cycle interpretation, and where the defining Hilbert-space domain ends. It does not fit parameters to primes or zeros, evaluate a Fredholm determinant, or claim a Riemann spectral identity.

Our contributions are:

1. a trace-class theorem for the frozen symmetric kernel on $\Re s > 1$, obtained from a Möbius rank-one decomposition;
2. an exact trace-power identity with primitive periods and repetitions, including closed formulas for periods one through three;
3. a target-free finite certificate using disjoint label blocks and exact rational arithmetic through $k = 6$;
4. a sharp operator-domain observation: the same counting-measure ℓ^2 matrix is not bounded for $\Re s \leq 1$.

The resulting Route-A tuple is

$$(A1_WEAK, A2_ANALYTIC_DETERMINANT, \\ A3_PARTIAL_ANALYTIC_STRUCTURE, A4_FAIL).$$

The next question is scalar continuation across the operator boundary. Any such continuation must be proved as a separate object-level theorem.

2 Frozen object and determinant convention

Definition 2.1 (Coprime renewal suspension). Let

$$\Sigma_{\text{cop}} = \{(n_k)_{k \in \mathbb{Z}} : n_k \in \{2, 3, \dots\}, \gcd(n_k, n_{k+1}) = 1 \text{ for every } k\}.$$

The left shift is the dynamics and the roof is $\tau((n_k)) = \log n_0$.

The edge potential is split symmetrically. On $\mathcal{H} = \ell^2(\{2, 3, \dots\})$, with counting measure, define

$$(L_s f)(m) = \sum_{\substack{n \in \{2, 3, \dots\} \\ \gcd(m, n) = 1}} (mn)^{-s/2} f(n), \quad \Re s > 1.$$

The real logarithm fixes the complex powers. A cyclic word $(n_0, \dots, n_{k-1})$ has edge product

$$\prod_{j=0}^{k-1} (n_j n_{j+1})^{-s/2} = \left(\prod_{j=0}^{k-1} n_j \right)^{-s}, \quad n_k = n_0.$$

This identity fixes the clock, normalization, and determinant data type.

Remark 2.2 (One ledger). The only object used below is

$$D_{\text{cop}}(s) = \det_F(I - L_s).$$

The reciprocal, logarithmic derivative, a scattering determinant, and the completed Riemann function are excluded from this stage. Trace class alone does not authorize a $\lambda = 1$ trace-log expansion unless a separate spectral-radius or norm bound is proved.

3 Trace-class theorem

Theorem 3.1 (Möbius nuclear decomposition). *For $\sigma = \Re s > 1$, L_s is trace class on $\mathcal{H}$, and $s \mapsto L_s$ is locally uniformly holomorphic in trace norm.*

Proof. Use

$$\mathbf{1}_{\gcd(m, n) = 1} = \sum_{d \mid m, n} \mu(d).$$

Set

$$a_d(m) = \mathbf{1}_{d \mid m} m^{-s/2}, \quad c_d(m) = \mathbf{1}_{d \mid m} m^{-\bar{s}/2}.$$

Then $L_s = \sum_{d \geq 1} \mu(d) a_d c_d^*$. The d -th term is rank one and

$$\|a_d c_d^*\|_1 = \|a_d\|_2 \|c_d\|_2 = S_d := \sum_{\substack{m \geq 2 \\ d \mid m}} m^{-\sigma}.$$

Since

$$\sum_{d \geq 1} |\mu(d)| S_d = \sum_{m \geq 2} 2^{\omega(m)} m^{-\sigma} = \frac{\zeta(\sigma)^2}{\zeta(2\sigma)} - 1 < \infty,$$

the rank-one series converges in trace norm. Local uniformity on compact subsets of $\Re s > 1$ follows from the same majorant with a compact lower bound for σ . Holomorphicity of each rank-one term gives the claim. $\square$

Corollary 3.2. *The Fredholm determinant $D_{\text{cop}}(s) = \det_F(I - L_s)$ is well-defined and holomorphic on $\Re s > 1$.*

Remark 3.3. The theorem is a half-plane statement. It does not assert that D_{cop} continues through $\Re s = 1$, nor does it identify a global divisor.

4 Exact trace powers and primitive cycles

Let P_N be the projection onto labels $2, \dots, N$. The finite matrices $P_N L_s P_N$ have the usual cyclic trace expansion. Trace-norm convergence passes this identity to the limit; alternatively, absolute values are bounded by $(\zeta(\sigma) - 1)^k$. Therefore, for every $k \geq 1$,

$$\text{Tr } L_s^k = \sum_{\substack{n_0, \dots, n_{k-1} \in \{2, 3, \dots\} \\ \gcd(n_j, n_{j+1}) = 1 \text{ cyclic}}} \left(\prod_{j=0}^{k-1} n_j \right)^{-s}.$$

If γ is a primitive directed cycle, write $d = |\gamma|$ and $w_\gamma = \prod_i n_i^{-s}$. Cyclic starting points give the exact ledger

$$\text{Tr } L_s^k = \sum_{\substack{\gamma \text{ primitive} \\ d|k}} d w_\gamma^{k/d}.$$

The factor is the primitive period d , not the total power k .

Proposition 4.1 (Periods one through three). $C_1(s) = 0$, and

$$C_2(s) = 2 \sum_{\substack{2 \leq a < b \\ \gcdop(a, b) = 1}} (ab)^{-s} = \frac{\zeta(s)^2}{\zeta(2s)} - 2\zeta(s) + 1.$$

Moreover,

$$C_3(s) = 6 \sum_{\substack{2 \leq a < b < c \\ \text{pairwise coprime}}} (abc)^{-s}.$$

Proof. A self-loop would require $\gcd(n, n) = 1$, impossible for $n \geq 2$. For period two, the two cyclic starting points represent one canonical rotation class with $a < b$. For period three, cyclic admissibility is pairwise coprimality, which forces distinct labels; six ordered triples represent two directed rotations for each unordered triple. $\square$

For a symbolic closed form, define

$$F_3(s) = \prod_p \frac{1 + 2p^{-s}}{1 - p^{-s}}.$$

Inclusion–exclusion of the forbidden label 1 gives

$$C_3(s) = F_3(s) - 3 \frac{\zeta(s)^2}{\zeta(2s)} + 3\zeta(s) - 1.$$

The Euler product is used only as a symbolic identity; no prime list is evaluated by the certificate.

5 The exact operator boundary

Proposition 5.1. *For $\sigma = \Re s \leq 1$, the frozen matrix L_s is not a bounded operator on $\ell^2(\{2, 3, \dots\})$.*

Proof. Apply L_s to e_2 . The only nonzero entries occur at odd $m \geq 3$, and

$$\|L_s e_2\|_2^2 = 2^{-\sigma} \sum_{\substack{m \geq 3 \\ m \text{ odd}}} m^{-\sigma}.$$

The odd subseries of the p-series diverges for $\sigma \leq 1$. □

This is a boundary for the frozen Hilbert-space realization, not a no-go theorem for every scalar continuation. A continuation of D_{cop} , a different Banach space, or a regularized operator would require a new source lock and a separate determinant identity. The current stage keeps all such objects outside scope.

6 Exact finite certificate

The generator uses only integer gcd tests and rational weights at $s = 2$. Validation uses labels $2, \dots, 10$; the sealed test uses the disjoint block $11, \dots, 18$. A separate repetition control uses labels $2, \dots, 8$ and checks powers $k = 1, \dots, 6$.

Table 1: Exact cyclic-word and primitive-cycle counts.

label block	period 1	period 2	period 3
validation 2–10	0	44	120
sealed test 11–18	0	40	132

For the validation block, the primitive cycle counts at periods two and three are 22 and 40, respectively. The exact $s = 2$ inclusion–exclusion checks return

$$C_2 = \frac{7591}{52920}, \quad C_3 = \frac{3637}{211680},$$

both by direct cyclic enumeration and by the finite formulas. Every row of the repetition ledger through $k = 6$ agrees as a reduced Fraction. These checks certify the combinatorial and determinant conventions; they do not measure agreement with any external spectrum.

7 Limitations and next theorem

The coprimality grammar is arithmetic-looking but is not a prime grammar. Nothing in the local transition rule singles out a primitive cycle for each prime, and no amplitude $\log p p^{-r/2}$ has been derived. The determinant is currently a half-plane Fredholm determinant, not a known entire function. Consequently there is no functional equation, Riemann–von Mangoldt counting law, completed- ξ divisor equality, natural quantization, or Route-B operator.

The next smallest verifiable task is to study scalar continuation or a same-object growth obstruction at $\Re s = 1$, while keeping the original ℓ^2 operator domain and determinant convention explicit. A successful continuation would need its own theorem; a failure would be a reusable candidate-local obstruction. Root searches and comparisons with Riemann zeros are deliberately deferred.

A Proof details

A.1 Trace-norm majorant

For $\sigma > 1$, the rank-one vectors in the Möbius decomposition satisfy

$$\|a_d\|_2^2 = \|c_d\|_2^2 = \sum_{\substack{m \geq 2 \\ d|m}} m^{-\sigma}.$$

The identity

$$\sum_{d \geq 1} |\mu(d)| \sum_{\substack{m \geq 2 \\ d|m}} m^{-\sigma} = \sum_{m \geq 2} 2^{\omega(m)} m^{-\sigma} = \frac{\zeta(\sigma)^2}{\zeta(2\sigma)} - 1$$

is absolutely convergent. It also supplies a compact-subset majorant, since σ has a positive margin above one on every compact subset of the defining half-plane.

A.2 Trace expansion

For a finite projection P_N , ordinary matrix multiplication gives

$$\mathrm{Tr}(P_N L_s P_N)^k = \sum_{\substack{n_0, \dots, n_{k-1} \leq N \\ \gcd(n_j, n_{j+1})=1 \text{ cyclic}}} \left(\prod_j n_j \right)^{-s}.$$

The trace-norm convergence of $P_N L_s P_N$ to L_s passes the trace of each fixed power to the limit. The same result follows directly from the absolute bound $(\zeta(\sigma) - 1)^k$, which permits Fubini and cyclic reindexing.

A.3 Primitive repetition factor

Every cyclic word has a unique minimal primitive period d dividing its length k . A primitive orbit of period d contributes exactly d cyclic starting points, and repeating it k/d times raises its weight to $w_\gamma^{k/d}$. Summing over primitive rotation classes gives the stated ledger.